Foundation Models for Fast AC-State Evaluation: A GridSFM-Augmented Optimal Power Shutoff Diagnostic Study

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Abstract

In this work, we embed the recently released GridSFM foundation model into the existing Optimal Power Shutoff problem as a testbed for foundation model transfer to a challenging topology-changing operating regime. On the GOC-500 bus grid example, we use GridSFM as a fast electrical-state predictor for externally generated line de-energization and load-service decisions and assess its behavior across wildfire-risk scenarios. We further fine-tune GridSFM using controlled variations in both training-set size and topology distribution to examine how adaptation affects predictive fidelity and downstream OPS performance. Fine-tuning improves state and physics metrics and produces closer agreement with fixed-decision AC-OPF reference solutions, while controlled warm-start experiments show only modest solver-speed benefits. The observed tradeoffs and findings establish a means for further studying foundation model integration in power-system decision-making.

1 Introduction

Optimal Power Shutoff and its computational challenge Wildfire mitigation creates a fundamental tradeoff: preventive line de-energization during Public Safety Power Shutoff (PSPS) events can reduce ignition exposure but also interrupt customers and reduce served demand [5]. The Optimal Power Shutoff (OPS) problem formalizes this as a topology-control problem, de-energizing high-risk lines while adjusting operational recourse, including generator redispatch, to preserve power delivery. The original OPS formulation uses linearized DC power flow [11], whereas higher-fidelity AC formulations better represent network operation at substantially greater computational cost as system size and candidate topologies increase [10]. For a fixed operating point, AC power flow solves nonlinear active- and reactive-power balance, commonly through iterative methods such as Newton–Raphson [12]; AC optimal power flow (AC-OPF) additionally optimizes operational controls subject to the nonlinear AC equations and operating constraints, often using methods such as primal–dual interior-point algorithms [13]. OPS compounds this cost through combinatorial topology search, since each candidate line-status decision induces a modified network requiring a new operating-state evaluation.

Surrogates and foundation-model application Learned PF/OPF surrogates can replace repeated nonlinear solves with direct state prediction, including topology-aware models for contingency analysis and changing network configurations [4, 7]. Because such models are often specialized to a grid, task, or operating distribution, recent work has pursued reusable neural solvers and grid foundation models designed for transfer across systems and conditions [8, 14]. GridSFM, studied here, is pretrained for AC-OPF across many grids and operating conditions and supports system-specific fine-tuning [14]. In wildfire-aware power systems, learning has instead primarily targeted ignition-risk estimation or discrete line-de-energization decisions [2, 6, 3], rather than prediction of the resulting AC operating state. Accordingly, using the OPS formulation of [11] as an operational

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reference, we integrate a pretrained AC-OPF foundation model as a surrogate AC-state predictor for externally generated topology and load-service decisions, which, to the best of our knowledge, is the **first such integration of a pretrained grid foundation model into the OPS problem**.

2 GridSFM-Augmented OPS Framework

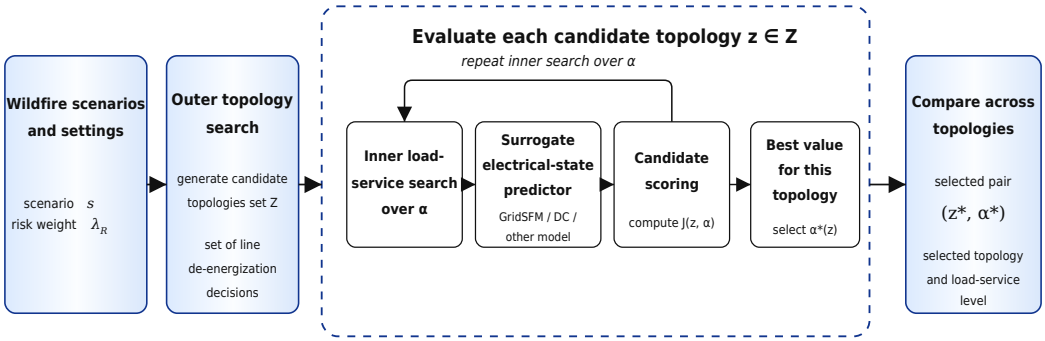


Figure 1: Overview of the fast enumeration framework, where the outer topology search first generates the candidate set $\mathcal{Z}$ and GridSFM-based continuous recourse is then evaluated for each $z \in \mathcal{Z}$.

Outer topology search The outer master generates a set $\mathcal{Z}$ of promising line de-energization topologies using low-cost wildfire-exposure and load-service proxies. For proxy risk weight λ_R^M,

$$\begin{aligned} &\min_{z} \quad \lambda_R^M \widehat{R}(z) + (1-\lambda_R^M)\widehat{L}(z) \quad \text{s.t.} \quad \sum_{\ell\in\mathcal{L}}(1-z_\ell) \le K, \quad z_\ell \in \{0,1\}, \quad K=2, \\ &\widehat{R}(z) = \frac{\sum_{\ell\in\mathcal{L}} w_\ell z_\ell}{\sum_{\ell\in\mathcal{L}} w_\ell}, \qquad\qquad \widehat{L}(z) = \sum_{\ell\in\mathcal{L}} c_\ell(1-z_\ell). \end{aligned} \tag{1}$$

Here, $z_\ell = 0$ and 1 denote de-energized and energized lines respectively, w_ℓ combines scenario wildfire exposure with baseline loading, and c_ℓ measures the served-demand impact of removing line ℓ. These quantities are precomputed, so candidate generation depends only on the binary topology vector z.

GridSFM continuous recourse For each $z \in \mathcal{Z}$, we optimize load service over the reduced space $\mathcal{A}_B(z)$ of five topology-relevant controls. Because GridSFM is used as a black-box electrical evaluator, we use derivative-free Powell search under $\mathcal{A}_B(z)$ for computationally tractable evaluation across multiple candidate topologies. With $\hat{x}_\theta = \mathrm{GridSFM}(z,\alpha)$, the best evaluated recourse is

$$\begin{aligned} &\alpha^*(z) \in \arg\min_{\alpha\in\mathcal{A}_B(z)} J(z,\alpha) = \lambda_R R_{\mathrm{norm}}(z,\hat{x}_\theta) + (1-\lambda_R)L_{\mathrm{shed}}(\alpha) + \rho_{\mathrm{phys}} P_{\mathrm{AC}}^{\mathrm{Total}}(\hat{x}_\theta), \\ &\text{s.t.} \quad 0 \le \alpha_i \le 1, \qquad i \in \mathcal{D}, \qquad \hat{x}_\theta = \mathrm{GridSFM}(z,\alpha). \end{aligned} \tag{2}$$

The objective combines baseline-normalized wildfire exposure over the AC transmission-line set $\mathcal{L}_{\mathrm{trans}}$, normalized unserved demand over load buses $\mathcal{D}$, and physics infeasibility:

$$\begin{aligned} R_{\mathrm{norm}}(z,\hat{x}_\theta) &= \frac{\sum_{\ell\in\mathcal{L}_{\mathrm{trans}}} z_\ell p_\ell^{\mathrm{env}} \mathrm{loading}_\ell^2(z,\hat{x}_\theta)}{\sum_{\ell\in\mathcal{L}_{\mathrm{trans}}} p_\ell^{\mathrm{env}} (\mathrm{loading}_\ell^{\mathrm{base}})^2}, \qquad L_{\mathrm{shed}}(\alpha) = \frac{\sum_{i\in\mathcal{D}} P_{d,i}^{\mathrm{base}}(1-\alpha_i)}{\sum_{i\in\mathcal{D}} P_{d,i}^{\mathrm{base}}}, \\ P_{\mathrm{AC}}^{\mathrm{Total}} &= P_{\mathrm{AC}}^{\mathrm{oper}} + P_{\mathrm{AC}}^{P/Q}. \end{aligned} \tag{3}$$

Here, R_{norm} compares wildfire-weighted post-decision loading against the intact baseline, L_{shed} measures the fraction of baseline demand not served with α_i denoting the served fraction at load bus i, and $P_{\mathrm{AC}}^{\mathrm{Total}}$ aggregates operational and active/reactive power-balance constraint violations, where Table 1 provides a representative decomposition.

After retaining $\alpha^*(z)$ for each candidate topology, the final OPS solution is selected across the evaluated $(z, \alpha^*(z))$ pairs. Full formulation details and search settings are provided in Appendix A.

3 Results

3.1 RQ1: GridSFM prediction performance in topology-changing OPS

Workflow and state response We first evaluate GridSFM on the diagnostic J-S1 setting, which contains one high-wildfire-risk line with relatively low expected service impact and is designed to test whether OPS decisions follow the intended risk pattern. At $\lambda_R = 0.8$, the solution opens the targeted high-risk branch 473 and branch 276 and reduces load 157 to $\alpha_{\text{eff}} = 0.0081$. Predicted voltages also remain within limits up to numerical tolerance. In this diagnostic setting, such interpretable state predictions provide a controlled proof of concept for using GridSFM inside OPS before considering larger, economically detailed wildfire-operation studies.

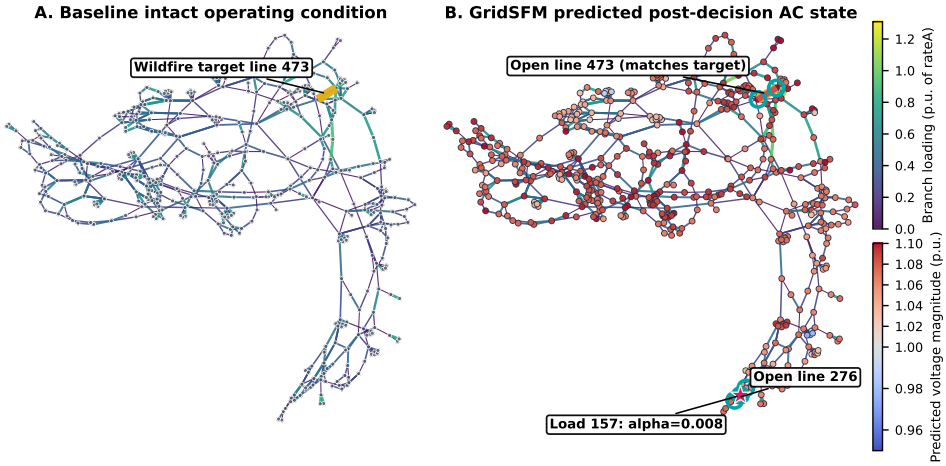


Figure 2: A depiction of the OPS workflow through the baseline operating condition and final OPS decision using the J-S1 wildfire scenario at $\lambda_R = 0.8$.

Physics consistency under topology shift Table 1 qualifies the preceding state prediction through explicit AC-physics constraint violations modeled in the objective function. Under the topology-changing regime, both operational violations and active/reactive power-balance residuals increase, indicating degraded physics consistency. This motivates a lightweight fine-tuning study in RQ2 to test whether limited system-specific adaptation can reduce these violations and improve stability across operating regimes while remaining easy to apply to new settings.

Table 1: Physics penalties for the J-S1 example.

Diagnostic	Intact	Modified
Voltage-limit penalty	0	0
Thermal-limit penalty	3.14×10^{-6}	1.58×10^{-4}
Generator-bound penalty	0	0
Source-less-service penalty	0	0
$\mathbf{P}_{\text{AC}}^{\text{oper}}$	$\mathbf{3.14\times10^{-6}}$	$\mathbf{1.58\times10^{-4}}$
Active-power balance MSE	0.0768	0.0887
Reactive-power balance MSE	0.0250	0.0792
$\mathbf{P}_{\text{AC}}^{\mathbf{P/Q}}$	**0.1018**	**0.1679**
$\mathbf{P}_{\text{AC}}^{\text{Total}}$	**0.1018**	**0.1681**

Candidate-state evaluation cost Across 54 unique fixed (z, α) instances, GridSFM is compared against the corresponding AC-OPF reference solved with IPOPT. GridSFM achieves median $3.35\times$ and $3.59\times$ speedups for the model inference itself and the complete candidate-to-state evaluation, including candidate-specific preparation and state extraction, respectively. Combined with the increased physics residuals under topology change, these results expose a clear speed–fidelity tradeoff: GridSFM substantially accelerates state evaluation, but improved physical fidelity remains necessary for reliable OPS assessment. The full table with the timing distributions and paired confidence intervals are displayed in Appendix B.

3.2 RQ2: Does fine-tuning improve GridSFM's physics fidelity?

Fine-tuning performance Model 1 is the released GridSFM checkpoint. Model 2 fine-tunes Model 1 on 1,000 GOC-500 full topology cases; Models 3 and 4 continue Model 2 with 500 additional full topology and N-1 samples, respectively. All checkpoints are evaluated on the same 375 FullTop and 375 N-1 held-out AC-OPF graphs and the same J-S1 physics-transfer pair: an unchanged *Intact* case and the selected *Modified* (z, α) decision from Figure 2.

Table 2: Fine-tuning evaluation and physics transfer.

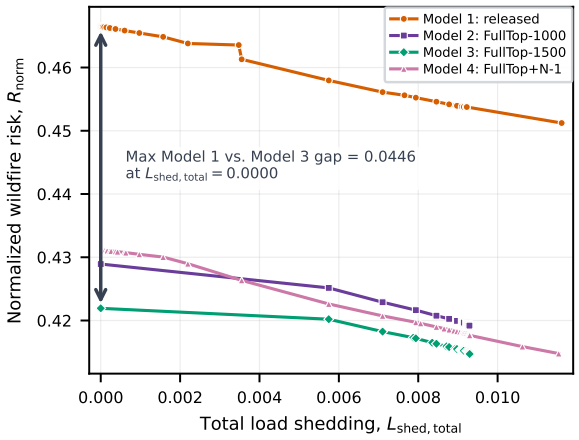
	FullTop test		N-1 test		$P_{\mathrm{AC}}^{\mathrm{oper}}$		$P_{\mathrm{AC}}^{P/Q}$	
Model	Loss	Br.-P MAE	Loss	Br.-P MAE	Intact	Modified	Intact	Modified
1	0.09330	0.07451	0.11847	0.07688	3.14×10^{-6}	1.58×10^{-4}	0.1018	0.1679
2	0.04079	0.04048	0.07015	0.04360	0	1.07×10^{-4}	0.0131	0.0738
3	**0.03811**	**0.03502**	0.06824	0.03869	0	1.09×10^{-4}	**0.0080**	0.0633
4	0.04038	0.03605	**0.05680**	**0.03804**	0	1.22×10^{-4}	0.0128	**0.0312**

Fine-tuning improves predictive and physics consistency relative to the released GridSFM checkpoint, although a structural shift remains between intact and topology-modified states (Table 2). N-1 fine-tuning provides the strongest modified-state physics consistency and a more stable regime transition, motivating broader topology-aware fine-tuning beyond this controlled setting. Full prediction metrics and held-out test details are provided in Appendix C.

3.3 RQ3: Do model improvements transfer to OPS behavior?

OPS frontier comparisons We compare Models 1–4 within the same GridSFM evaluator and search settings. Fine-tuning shifts the empirical wildfire-risk/load-service frontiers toward lower objective values (Figure 3), with the larger full topology adaptation in Model 3 providing the strongest performance. Notably, Model 4 achieves the lowest modified-state AC-physics residual after N-1 adaptation, yet does not outperform Model 3 on the OPS frontier. This distinction motivates further study of adaptation-set size versus topology-aware sampling beyond the controlled N-1 setting and their effects on downstream operational decisions.

Figure 3: GridSFM evaluated frontiers on J-S1.



Downstream transfer to AC-OPF and warm starts For each selected OPS decision, we compute the nonlinear fixed-decision AC-OPF reference and measure disagreement with the GridSFM evaluation. Comparing Model 1 and 3, state NRMSE, wildfire-risk discrepancy, and objective discrepancy decrease by approximately 42%, 76%, and 83%, respectively, showing that improved GridSFM fidelity transfers to closer agreement with the nonlinear AC evaluation. We additionally test whether GridSFM-predicted states improve initialization of the same AC-OPF solves. Across 75 fixed instances, informed starts provide only modest changes in IPOPT iterations and solve time, with no monotonic improvement as GridSFM prediction fidelity increases. Thus, improved state accuracy transfers strongly to AC-OPF agreement but does not substantially accelerate the nonlinear AC-OPF solve itself. Full frontier comparisons, Reference A/B, and warm-start results are in Appendix D.

4 Conclusion and Outlook

Scope and findings This controlled diagnostic study integrates a pretrained grid foundation model into topology-changing OPS as a surrogate AC-state predictor. Supporting interpretable evaluation, we restrict the study to constructed wildfire scenarios, a shutoff budget of $K \leq 2$, and a bounded five-control load-service search. GridSFM achieves meaningful speedups, while fine-tuning reduces AC-physics residuals and improves agreement with the fixed-decision AC-OPF reference. A structural shift nevertheless remains between intact and topology-modified states, and the results establish a proof of concept rather than a deployment-ready OPS policy.

Outlook Future work should move beyond bounded enumeration toward gray-box optimization that embeds foundation-model states and derivatives directly into joint topology selection and continuous recourse. Security-constrained OPS can test robustness to subsequent contingencies [9], while larger systems, realistic wildfire-risk data, and explicit economic and reliability objectives can assess the operational value of the observed speed–fidelity tradeoff. Broader topology-aware adaptation should distinguish training-set size from topology diversity and evaluate their downstream effects.

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A Full OPS framework details

This appendix expands the GridSFM-augmented OPS framework introduced in the main paper. The method is sequential: an inexpensive outer topology search first generates a finite candidate set $\mathcal{Z}$, after which GridSFM state prediction and bounded continuous recourse are evaluated separately for each candidate topology. Appendix A.1 provides the full formulation, while Appendix A.2 specifies the controlled search settings used in this study.

A.1 Formulation

A.1.1 Outer topology search

Transmission-line topology decision Let $\mathcal{L}_{\text{phys}}$ denote the physical branch set represented in the electrical network, partitioned as

$$\mathcal{L}_{\text{phys}} = \mathcal{L}_{\text{trans}} \cup \mathcal{L}_{\text{tr}}, \tag{4}$$

where $\mathcal{L}_{\text{trans}}$ is the full set of physical AC transmission lines represented by GridSFM's `ac_line` family and $\mathcal{L}_{\text{tr}}$ contains transformer-family branches. Transformer branches remain electrically represented but are excluded from wildfire exposure and preventive line-de-energization decisions.

For each $\ell \in \mathcal{L}_{\text{trans}}$, define

$$z_\ell = \begin{cases} 1, & \text{line } \ell \text{ is energized,} \\ 0, & \text{line } \ell \text{ is de-energized.} \end{cases} \tag{5}$$

Thus, $z = (z_\ell)_{\ell \in \mathcal{L}_{\text{trans}}}$ defines the controllable AC-transmission topology.

Each AC transmission line also receives an exogenous environmental wildfire-exposure score

$$p_\ell^{\text{env}} \geq 0, \qquad \ell \in \mathcal{L}_{\text{trans}}. \tag{6}$$

In this controlled study these values are prescribed by the diagnostic wildfire scenario. More generally, p_ℓ^{env} may be supplied by an external statistical or physics-based ignition-risk model incorporating environmental and asset information such as wind, temperature, vegetation, or equipment characteristics.

Outer proxy problem For proxy wildfire-risk weight $\lambda_R^M \in [0, 1]$, the outer topology problem is

$$\begin{aligned} \min_{z} \quad & \lambda_R^M \widehat{R}(z) + (1 - \lambda_R^M)\widehat{L}(z) \\ \text{s.t.} \quad & \sum_{\ell \in \mathcal{L}_{\text{trans}}} (1 - z_\ell) \leq K, \qquad z_\ell \in \{0, 1\}. \end{aligned} \tag{7}$$

The two topology-dependent proxy terms are

$$\widehat{R}(z) = \frac{\sum_{\ell \in \mathcal{L}_{\text{trans}}} w_\ell z_\ell}{\sum_{\ell \in \mathcal{L}_{\text{trans}}} w_\ell}, \qquad \widehat{L}(z) = \sum_{\ell \in \mathcal{L}_{\text{trans}}} c_\ell (1 - z_\ell). \tag{8}$$

The precomputed wildfire proxy weight is

$$w_\ell = p_\ell^{\text{env}} \left(\text{loading}_\ell^{\text{base}} \right)^2, \tag{9}$$

where $\text{loading}_\ell^{\text{base}}$ is the stored loading ratio under the intact AC operating condition. Hence, $\widehat{R}(z)$ measures the fraction of baseline-weighted wildfire exposure remaining after the topology decision without requiring GridSFM inference or a new nonlinear AC solve.

The service-consequence score $c_\ell \geq 0$ measures the estimated loss of served active demand after removing line ℓ alone:

$$c_\ell = \frac{\max\left\{0,\ S_D^{\text{base}} - S_D^{(-\ell)}\right\}}{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}}}, \tag{10}$$

where S_D^{base} is served active demand in the intact network and $S_D^{(-\ell)}$ is served demand after the single-line removal. The score c_ℓ is used only to guide topology generation and is not included in the post-decision wildfire-exposure calculation.

The outer proxy master is implemented in Gurobi and repeatedly solved to construct the candidate topology set $\mathcal{Z}$. Its topology and evaluation budgets are specified in Appendix A.2.

A.1.2 GridSFM state prediction and continuous recourse

Load-service decision Let $\mathcal{N}$ denote the bus set, $\mathcal{D} \subseteq \mathcal{N}$ the load buses, $\mathcal{G}$ the generator set, and $\mathcal{G}_i \subseteq \mathcal{G}$ the generators located at bus i. For a fixed topology $z \in \mathcal{Z}$, let $\mathcal{D}_{\mathrm{cmd}}(z) \subseteq \mathcal{D}$ denote the load buses available to bounded continuous recourse. For each selected bus,

$$0 \le \alpha_i \le 1, \qquad i \in \mathcal{D}_{\mathrm{cmd}}(z), \tag{11}$$

where α_i denotes the requested fraction of baseline demand served.

Let $\mathcal{I}_{\mathrm{SL}}(z)$ contain load buses in components disconnected from every source bus under topology z. Effective load service is

$$\alpha_i^{\mathrm{eff}}(z,\alpha) = \begin{cases} 0, & i \in \mathcal{I}_{\mathrm{SL}}(z), \\ \alpha_i, & i \in \mathcal{D}_{\mathrm{cmd}}(z) \setminus \mathcal{I}_{\mathrm{SL}}(z), \\ 1, & i \in \mathcal{D} \setminus \left(\mathcal{D}_{\mathrm{cmd}}(z) \cup \mathcal{I}_{\mathrm{SL}}(z)\right). \end{cases} \tag{12}$$

Thus, source-less islanding takes precedence over any requested control; a connected selected load follows α_i, and a connected non-selected load remains at full baseline service.

GridSFM electrical-state prediction For a fixed topology and effective load-service decision, GridSFM predicts the post-decision electrical state

$$\hat{x}_\theta = \mathrm{GridSFM}\left(z, \alpha^{\mathrm{eff}}\right). \tag{13}$$

The electrical-state quantities used in this study are represented by

$$\hat{x}_\theta = \left(\hat{\theta},\, \hat{V},\, \hat{P}_g,\, \hat{Q}_g,\, \hat{P}_{ij},\, \hat{Q}_{ij},\, \hat{P}_{ji},\, \hat{Q}_{ji}\right), \tag{14}$$

together with GridSFM's feasibility output. These quantities provide the voltage, generation, and two-ended branch-flow information used to evaluate the modified AC operating state.

Continuous recourse problem For each topology $z \in \mathcal{Z}$, the best evaluated load-service decision is

$$\begin{aligned} \alpha^*(z) \in \arg\min_{\alpha \in \mathcal{A}_B(z)} \quad & J(z,\alpha) = \lambda_R R_{\mathrm{norm}}(z,\hat{x}_\theta) + (1-\lambda_R) L_{\mathrm{shed}}(z,\alpha) + \rho_{\mathrm{phys}} P_{\mathrm{AC}}^{\mathrm{Total}}(\hat{x}_\theta) \\ \text{s.t.} \quad & 0 \le \alpha_i \le 1, \qquad i \in \mathcal{D}_{\mathrm{cmd}}(z), \qquad \hat{x}_\theta = \mathrm{GridSFM}(z, \alpha^{\mathrm{eff}}). \end{aligned} \tag{15}$$

Here $\lambda_R \in [0,1]$ controls the wildfire-exposure versus load-service tradeoff and $\rho_{\mathrm{phys}} \ge 0$ weights predicted AC-physics infeasibility. The bounded set $\mathcal{A}_B(z)$ is searched using `scipy.optimize.minimize` with the derivative-free Powell method, with the controlled search settings specified in Appendix A.2.

Post-decision wildfire exposure Unlike the outer proxy, the recourse objective uses GridSFM-predicted AC line flows under each evaluated candidate. The normalized wildfire exposure is

$$R_{\mathrm{norm}}(z,\hat{x}_\theta) = \frac{R_{\mathrm{current}}(z,\hat{x}_\theta)}{R_{\mathrm{base}}}. \tag{16}$$

For an AC transmission line $\ell = (i,j)$ with thermal rating $S_\ell^{\max}$, its post-decision loading ratio is

$$\mathrm{loading}_\ell(z,\hat{x}_\theta) = \frac{\max\left\{\sqrt{\hat{P}_{ij}^2 + \hat{Q}_{ij}^2}, \sqrt{\hat{P}_{ji}^2 + \hat{Q}_{ji}^2}\right\}}{S_\ell^{\max}}. \tag{17}$$

The corresponding baseline quantity $\mathrm{loading}_\ell^{\mathrm{base}}$ is computed using the same expression, replacing the GridSFM-predicted branch flows with the stored two-ended active and reactive branch flows from the intact baseline operating condition before any topology change.

The current and baseline wildfire exposures are

$$\begin{aligned} R_{\mathrm{current}}(z,\hat{x}_\theta) &= \sum_{\ell \in \mathcal{L}_{\mathrm{trans}}} z_\ell p_\ell^{\mathrm{env}} \left[\mathrm{loading}_\ell(z,\hat{x}_\theta)\right]^2, \\ R_{\mathrm{base}} &= \sum_{\ell \in \mathcal{L}_{\mathrm{trans}}} p_\ell^{\mathrm{env}} \left(\mathrm{loading}_\ell^{\mathrm{base}}\right)^2. \end{aligned} \tag{18}$$

Thus, R_{current} uses candidate-specific GridSFM flows, whereas R_{base} uses the intact baseline flows. Consequently, $R_{\text{norm}} < 1$ indicates lower wildfire-weighted line loading than the baseline and $R_{\text{norm}} > 1$ indicates greater exposure.

Normalized load shedding Normalized unserved active demand is

$$L_{\text{shed}}(z, \alpha) = \frac{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}} \left(1 - \alpha_i^{\text{eff}}(z, \alpha)\right)}{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}}}. \tag{19}$$

This term captures both intentional load-service reduction at controlled buses and involuntary loss of service caused by source-less islanding.

Physics-infeasibility decomposition GridSFM predictions are qualified through

$$P_{\text{AC}}^{\text{Total}} = w_{\text{oper}} P_{\text{AC}}^{\text{oper}} + w_{P/Q} P_{\text{AC}}^{P/Q}, \tag{20}$$

where the reported experiments use unit group weights. The operational component measures operating-limit violations, while $P_{\text{AC}}^{P/Q}$ measures nonlinear active- and reactive-power balance residuals.

The operational component is

$$P_{\text{AC}}^{\text{oper}} = P_{\text{AC}}^{V} + P_{\text{AC}}^{\text{thermal}} + P_{\text{AC}}^{G} + P_{\text{AC}}^{\text{island}}. \tag{21}$$

Voltage-limit violations are

$$P_{\text{AC}}^{V} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \left[\max(0, V_i^{\min} - \hat{V}_i)^2 + \max(0, \hat{V}_i - V_i^{\max})^2 \right]. \tag{22}$$

For the energized transmission-line set $\mathcal{L}_{\text{active}}(z) = \{\ell \in \mathcal{L}_{\text{trans}} : z_\ell = 1\}$, thermal violation is

$$P_{\text{AC}}^{\text{thermal}} = \frac{1}{|\mathcal{L}_{\text{active}}(z)|} \sum_{\ell \in \mathcal{L}_{\text{active}}(z)} \max\left[0,\, \text{loading}_\ell(z, \hat{x}_\theta) - 1\right]^2. \tag{23}$$

The normalized active-generator bound penalty is

$$P_{\text{AC}}^{G} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \left[\max\left(0, \frac{\hat{P}_g - P_g^{\max}}{P_g^{\max} - P_g^{\min} + \epsilon}\right)^2 + \max\left(0, \frac{P_g^{\min} - \hat{P}_g}{P_g^{\max} - P_g^{\min} + \epsilon}\right)^2 \right], \tag{24}$$

where $\epsilon > 0$ prevents division by zero.

The source-less-service diagnostic is

$$P_{\text{AC}}^{\text{island}} = \frac{\sum_{i \in \mathcal{I}_{\text{SL}}(z) \cap \mathcal{D}} P_{d,i}^{\text{base}} \left(\alpha_i^{\text{eff}}\right)^2}{\sum_{i \in \mathcal{D}} P_{d,i}^{\text{base}}}. \tag{25}$$

Because source-less loads are assigned zero effective service, this term also serves as a diagnostic of island-handling consistency.

For each bus i, the active- and reactive-power balance residuals are

$$\begin{aligned} r_i^P &= \sum_{g \in \mathcal{G}_i} \hat{P}_g - \alpha_i^{\text{eff}} P_{d,i}^{\text{base}} - P_i^{\text{AC}}(\hat{V}, \hat{\theta}, z), \\ r_i^Q &= \sum_{g \in \mathcal{G}_i} \hat{Q}_g - \alpha_i^{\text{eff}} Q_{d,i}^{\text{base}} - Q_i^{\text{AC}}(\hat{V}, \hat{\theta}, z). \end{aligned} \tag{26}$$

Using system power base S^{base},

$$P_{\text{AC}}^{P} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \left(\frac{r_i^P}{S^{\text{base}}} \right)^2, \qquad P_{\text{AC}}^{Q} = \frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} \left(\frac{r_i^Q}{S^{\text{base}}} \right)^2, \tag{27}$$

with

$$P_{\text{AC}}^{P/Q} = P_{\text{AC}}^{P} + P_{\text{AC}}^{Q}. \tag{28}$$

A.2 Search settings

The following settings instantiate the formulation above for the controlled GOC-500 study. They were selected to support repeated multi-topology evaluation on a laptop-scale environment and are experimental controls rather than intrinsic restrictions of the framework.

A.2.1 Outer topology search settings

Risk weighting The primary experiments use the same wildfire-risk preference for topology generation and continuous recourse:

$$\lambda_R^M = \lambda_R. \tag{29}$$

The two weights remain conceptually distinct: λ_R^M acts on the precomputed proxy quantities $\widehat{R}(z)$ and $\widehat{L}(z)$, while λ_R acts on the GridSFM-evaluated post-decision quantities.

The stages may also be decoupled. In particular,

$$\lambda_R^M = 1, \qquad \lambda_R = 0, \tag{30}$$

prioritizes wildfire-exposure reduction during topology generation and load delivery during continuous recourse, yielding the topology-first maximum-load-delivery-style specialization associated with the original OPS formulation [11].

Topology budgets The reported study uses

$$K = 2, \qquad B_Z = 100. \tag{31}$$

Thus, each topology may de-energize at most two AC transmission lines, and the outer search generates at most 100 distinct candidates for each search configuration. These values were selected for the controlled laptop-scale study; K and B_Z can be independently increased, and the topology cardinality restriction can be removed in broader studies.

The proxy master is implemented in Gurobi and solved repeatedly. After obtaining topology $z^{(r)}$, the no-good cut

$$\sum_{\ell : z_\ell^{(r)} = 1} (1 - z_\ell) + \sum_{\ell : z_\ell^{(r)} = 0} z_\ell \geq 1 \tag{32}$$

excludes exactly that binary line-status vector before the next solve. Repeated optimal solves therefore generate a ranked pool of up to 100 distinct proxy topologies rather than repeatedly returning the same master solution.

A.2.2 Continuous recourse search settings

Topology-relevant load selection The bounded recourse study uses

$$q = 5 \tag{33}$$

load-service coordinates for each topology. After the proposed lines are removed from the physical network graph, all endpoints of the de-energized lines are treated as simultaneous distance origins. For each eligible load bus,

$$d_i(z) = \min_{v \in \mathcal{E}_{\text{off}}(z)} d_{\text{graph}}(i, v), \tag{34}$$

where $\mathcal{E}_{\text{off}}(z)$ contains the shutoff-line endpoints and d_{graph} is unweighted shortest-path distance in the post-outage network.

Source-less loads are excluded because their effective service is already forced to zero. Remaining loads are ranked lexicographically by

$$\left(d_i(z),\ -P_{d,i}^{\text{base}},\ \text{ID}_i\right). \tag{35}$$

Network proximity to the nearest de-energized-line endpoint is therefore the primary criterion. Among loads at the same distance, larger baseline active demand is prioritized, with load ID used for deterministic tie-breaking. The first five loads define $\mathcal{D}_{\text{cmd}}(z)$; there is no intermediate nearest-neighbor pool.

Continuous evaluation budget For every $z \in \mathcal{Z}$, the bounded load-service problem is solved using the Powell method through `scipy.optimize.minimize`. GridSFM is treated as a black-box state predictor, so no foundation-model derivatives are required.

The study permits at most

$$B_\alpha = 20 \tag{36}$$

continuous objective evaluations per topology. Each evaluation proposes a load-service vector, obtains the corresponding GridSFM state, and evaluates $J(z, \alpha)$. The controlled search settings are therefore

$$K = 2, \qquad B_Z = 100, \qquad q = 5, \qquad B_\alpha = 20. \tag{37}$$

These settings can be independently expanded when additional computational resources or a larger decision space are desired.

B Speedups against AC-OPF reference

Table 3: Base released GridSFM candidate-state evaluation timing.

Timing boundary	**GridSFM median [IQR]**	**AC-OPF/IPOPT median [IQR]**	**Paired median speedup [95% CI]**
Model inference	0.222 [0.220–0.224] s	0.746 [0.719–0.777] s	3.35× [3.31–3.45]
Complete state evaluation	0.297 [0.293–0.305] s	1.092 [1.012–1.116] s	3.59× [3.48–3.68]

GridSFM is compared against the corresponding AC-OPF reference solved with IPOPT across 54 unique fixed (z, α) instances. The model inference boundary compares GridSFM inference itself with the IPOPT AC-OPF solve, while complete state evaluation additionally includes candidate-specific preparation and state extraction. Times report median [IQR], and speedups report paired medians with 95% bootstrap confidence intervals.

C Fine-tuning evaluation

This appendix provides additional details for the fine-tuning evaluation summarized in Table 2. The reported results combine a sealed FullTop/N-1 AC-OPF prediction test with a separate matched J-S1 physics-transfer evaluation.

The held-out prediction test uses solved GOC-500 AC-OPF cases based on the PGLib-OPF system [1]. The sealed test set contains 375 FullTop cases with the complete intact topology and 375 N-1 cases containing a single transmission-element contingency. Training, validation, and test graph identities are disjoint, and every checkpoint is evaluated without further training on the same ordered set of 750 held-out graphs.

The physics-transfer test uses the same representative J-S1 operating conditions for every checkpoint. The *Intact* condition uses the all-energized topology with full baseline load service, while the *Modified* condition applies the common OPS decision opening branches 276 and 473 and setting $\alpha_{157} = 0.00813062$. Holding this intervention fixed isolates differences in the GridSFM-predicted electrical states across checkpoints.

Evaluation metrics *Loss* is GridSFM's composite evaluation loss, which includes its state-prediction, generation-cost, feasibility, power-balance, branch-flow, and thermal evaluation terms. It is distinct from the wildfire-aware OPS objective $J(z, \alpha)$. *Cost MAPE* measures generation-cost percentage error, *Br.-P MAE* measures mean absolute error in active branch-flow prediction, *P-KCL residual* measures active-power balance consistency, and *Feas. acc.* reports feasibility-classification accuracy.

For the matched J-S1 evaluation, $P_{\mathrm{AC}}^{\mathrm{oper}}$ is the operational-constraint penalty and $P_{\mathrm{AC}}^{P/Q}$ is the combined active- and reactive-power balance residual defined in Appendix A.1.

Table 4: Full held-out GridSFM evaluation on the common FullTop and N-1 test sets. Each test contains 375 AC-OPF cases.

Test	Model	Loss	Cost MAPE	Br.-P MAE	P-KCL residual	Feas. acc.
FullTop	Model 1	0.093298	0.008437	0.074514	6.656×10^{-4}	1.000000
FullTop	Model 2	0.040794	0.007164	0.040482	2.631×10^{-4}	1.000000
FullTop	Model 3	**0.038112**	0.005520	**0.035024**	2.209×10^{-4}	1.000000
FullTop	Model 4	0.040381	**0.005228**	0.036046	2.398×10^{-4}	1.000000
N-1	Model 1	0.118472	0.011977	0.076878	6.742×10^{-4}	0.997333
N-1	Model 2	0.070146	0.008770	0.043597	2.924×10^{-4}	0.992000
N-1	Model 3	0.068235	0.007789	0.038692	2.578×10^{-4}	0.992000
N-1	Model 4	**0.056797**	**0.006347**	**0.038042**	2.550×10^{-4}	**1.000000**

D OPS model results

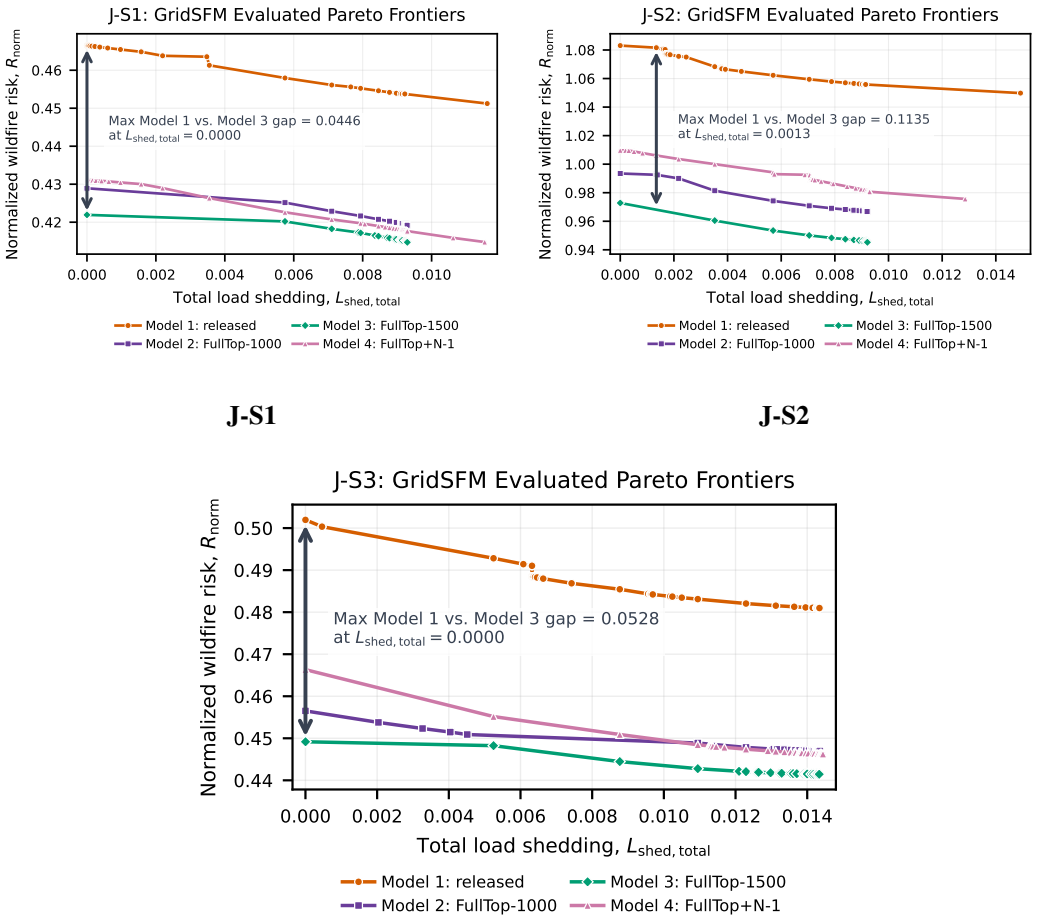
D.1 Full frontier comparisons

The three diagnostic scenarios use the same GOC-500 electrical network and baseline operating condition while varying the imposed wildfire-exposure pattern. They are controlled tests designed to produce interpretable risk patterns rather than calibrated estimates of real wildfire probabilities.

Table 5: Controlled wildfire scenarios used in the Stage J OPS study

Scenario	Diagnostic construction	Intended test
J-S1	High-risk line with relatively low service consequence	Whether the targeted high-risk line is removed when its service consequence is modest
J-S2	High-risk line with high service consequence	Whether the framework responds to a direct conflict between wildfire exposure and load service
J-S3	High-risk line in an electrically redundant alternate-path setting	Whether network redundancy enables exposure reduction with limited forced load loss

All three scenarios are evaluated using the common topology and continuous recourse settings defined in Appendix A.2. Across these diagnostics, Model 3 provides the strongest evaluated wildfire-risk/load-service frontiers among Models 1–4, consistent with the J-S1 result emphasized in the main paper. For each model, scenario, and risk weight, the framework ultimately produces a selected (z^*, α^*) decision; these selected decisions provide the common inputs to the nonlinear AC-reference studies below.



J-S1 **J-S2**

J-S3

Figure 4: Complete GridSFM-evaluated wildfire-risk/load-service frontiers for the three controlled diagnostic scenarios

D.2 Reference A and B

Guided-DC and GridSFM cannot be compared through their native electrical states or native Pareto objectives directly. GridSFM predicts an AC state containing active and reactive quantities and uses apparent-power loading, whereas Guided-DC represents only the corresponding DC quantities and uses active-power loading. Instead, each evaluator contributes its final selected (z^*, α^*) decision, which is then passed unchanged to a common nonlinear AC reference. This provides a consistent AC basis for comparing decisions selected by Guided-DC and Models 1–4.

D.2.1 Reference A: fixed-decision economic AC-OPF

For each model/scenario/risk-weight combination, Reference A fixes the selected OPS decision (z^*, α^*) and solves the corresponding nonlinear AC-OPF using PowerModels' `ACPPowerModel` with IPOPT.

Reference A therefore provides a common AC operating state and economic dispatch for comparing the decisions selected by Guided-DC and Models 1–4.

Across the controlled study, decisions selected by the GridSFM evaluators generally lead to lower Reference A economic AC-OPF costs than Guided-DC once wildfire risk receives positive weight, with Model 3 among the strongest families across the tested settings. This comparison concerns the

economic dispatch associated with the selected (z^*, α^*) decisions; the economic objective itself is not part of the native wildfire-aware OPS merit.

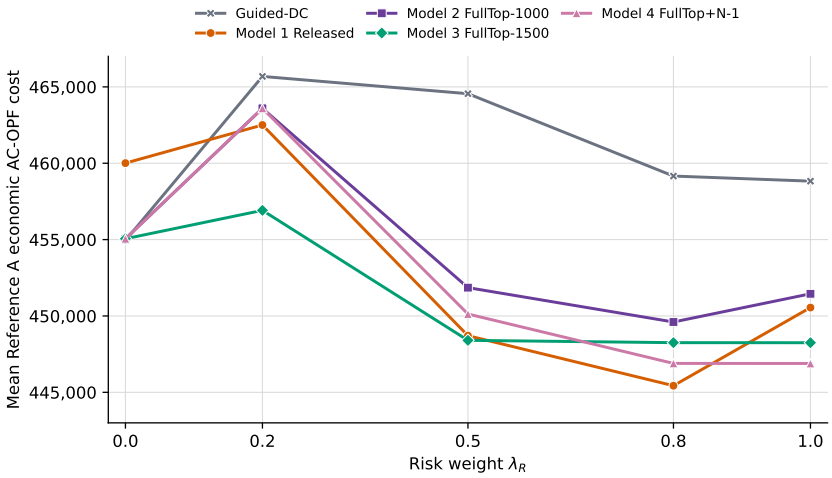


Figure 5: Mean Reference A economic AC-OPF cost associated with the selected OPS decisions across risk weights

The converged Reference A state also provides a common state-level comparison with the electrical state produced by each native evaluator.

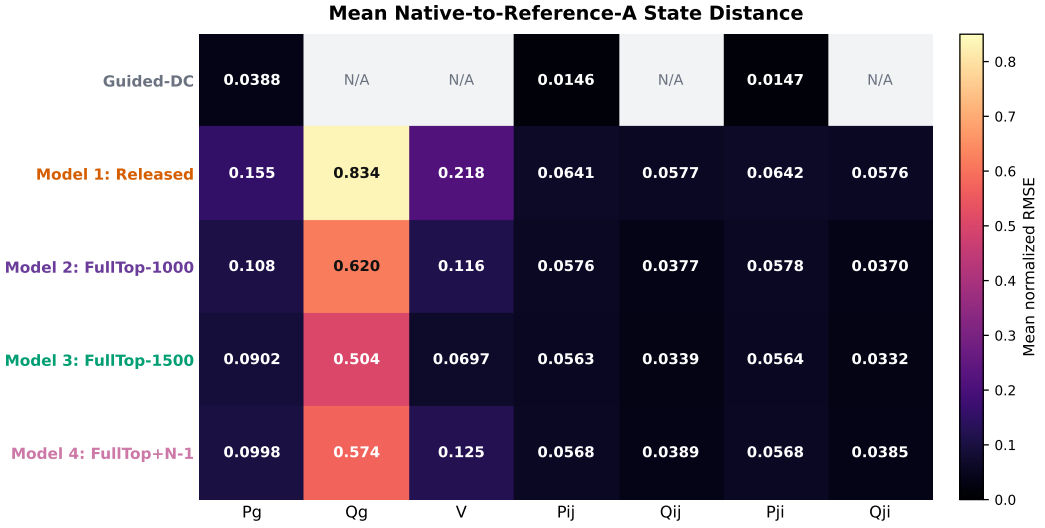


Figure 6: Mean native-to-Reference-A state discrepancy for the selected OPS decisions

Guided-DC does not represent reactive generation, voltage magnitude, or reactive branch-flow quantities, so those state-distance entries are unavailable rather than zero.

D.2.2 Reference B: fixed-topology maximum load delivery

Reference B fixes only the topology selected by each finalist,

$$z = z^*, \tag{38}$$

and releases the native load-service decision. It then solves the nonlinear AC maximum-load-delivery problem

$$\max_{\alpha,\, x^{\mathrm{AC}}} \quad \sum_{i \in \mathcal{D}} \alpha_i P_{d,i}^{\mathrm{base}} \tag{39}$$

subject to the fixed topology, nonlinear AC active/reactive power balance, proportional active/reactive load service, voltage and generator limits, thermal limits, and source-less-island treatment.

Reference B therefore measures the maximum load that the selected topology can support under nonlinear AC physics. Comparing this value with the service selected by the native evaluator distinguishes load shedding chosen by the surrogate search from load shedding that is unavoidable because of the topology itself. We report the service gap

$$\Delta S^B = S^B_{\max} - S_{\text{selected}}. \tag{40}$$

Because Reference B depends on the selected topology rather than the native AC state, Guided-DC can be evaluated through the same formulation as every GridSFM model.

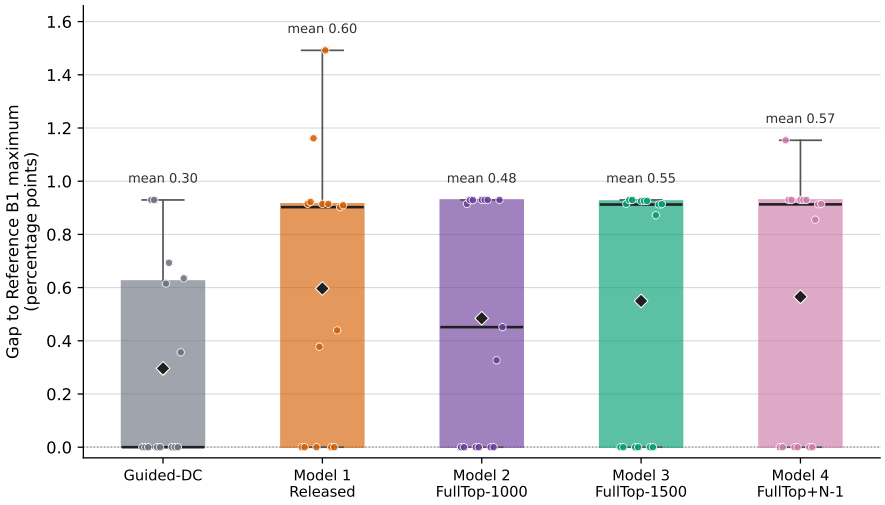


Figure 7: Gap between native selected service and the Reference B maximum AC-deliverable service

D.3 Warm-start results

The warm-start experiment separates two distinct roles: *finalist family* and *initializer*. The finalist family determines the selected (z^*, α^*) decision and therefore which fixed nonlinear Reference A AC-OPF instance IPOPT must solve. The initializer independently determines the electrical state supplied to IPOPT before solving that same instance.

Five finalist families are evaluated across three diagnostic scenarios and five risk weights, producing

$$5 \times 3 \times 5 = 75 \tag{41}$$

fixed (z^*, α^*) Reference A instances. Each instance is then solved from seven different initializations, giving

$$75 \times 7 = 525 \tag{42}$$

successful IPOPT solves. This crossed construction allows the effect of the selected OPS decision to be distinguished from the effect of the initial electrical state.

Initialization policies The seven initializers are the cold start, Guided-DC, Models 1–4, and the exact Reference A primal state. The cold start uses

$$V_i = 1, \qquad \theta_i = 0, \qquad P_g = \frac{P_g^{\min} + P_g^{\max}}{2}, \qquad Q_g = 0. \tag{43}$$

Guided-DC supplies P_g and voltage angle while retaining $V = 1$ and $Q_g = 0$. Each GridSFM initializer supplies P_g, Q_g, V, and θ from its predicted state. Exact A supplies only the converged primal AC state, providing an upper-reference initialization rather than a full IPOPT restart.

The study measures warm starts of the fixed Reference A AC-OPF solve, not of the full OPS procedure. Timing uses IPOPT-reported solve time and excludes model inference, state loading, surrogate scoring, and publication I/O.

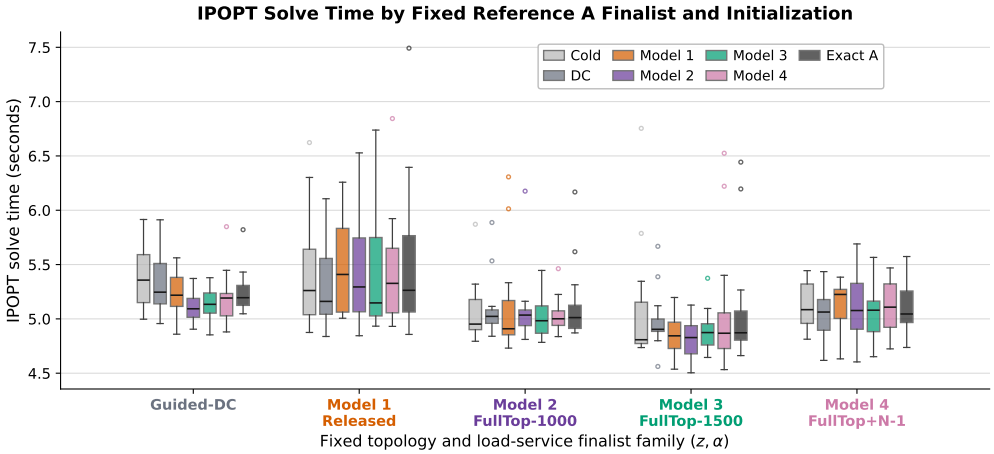


Figure 8: IPOPT solve-time distributions crossed by finalist family and initialization policy

Finalist-family aggregation Each finalist-family row aggregates the 15 selected (z^*, α^*) decisions from three scenarios and five risk weights under all seven initializers, yielding 105 solves per row. Because initialization is averaged over within each family, this comparison primarily measures the nonlinear difficulty of the AC-OPF instances induced by each model's selected OPS decisions.

Table 6: Reference A solve statistics grouped by finalist family

Finalist family	Mean time (s)	Median (s)	Mean iterations
Model 3: FullTop-1500	**4.963**	**4.872**	**30.99**
Model 2: FullTop-1000	5.074	5.001	31.70
Model 4: FullTop+N-1	5.096	5.080	32.98
Guided-DC	5.240	5.195	32.37
Model 1: Released	5.427	5.281	34.21

Model 3 has the lowest mean solve time, median solve time, and mean iteration count under the finalist-family aggregation. Thus, among the tested families, the (z^*, α^*) decisions selected by Model 3 produce Reference A instances that are, on average, the easiest for IPOPT to solve across the common set of initializers.

Initializer aggregation The complementary aggregation instead holds the 75 fixed (z^*, α^*) instances common and groups results by the state used to initialize IPOPT. Each row therefore contains one solve for every finalist, scenario, and risk-weight combination and directly measures initializer performance.

Table 7: Reference A solve statistics grouped by IPOPT initializer

IPOPT initializer	Mean time (s)	Median (s)	Mean iterations
Model 3 full state	**5.104**	5.055	32.49
Model 2 full state	5.108	**5.039**	32.45
DC partial state	5.151	5.075	**31.57**
Model 4 full state	5.159	5.067	32.32
Model 1 full state	5.160	5.104	32.25
Exact A primal state	5.216	5.089	32.43
Cold start	5.221	5.113	33.64

No single initializer dominates: Model 3 has the lowest mean solve time, Model 2 the lowest median, and Guided-DC the fewest mean iterations. Warm-start gains are therefore modest and non-monotonic with GridSFM fidelity; Model 3's stronger finalist-family result reflects the (z^*, α^*) instances it selects, not a uniquely superior initialization.

AI4PowerGrids domain evaluation checklist

1. **Physics feasibility.** Yes. We explicitly provide AC constraint violation statistics throughout this study. The total physics infeasibility is presented as two subterms of operational and active/reactive power-balance residuals computed from the full nonlinear AC equations. This can be seen in our Table 1 through a representative instance of the released GridSFM checkpoint. Additionally, Table 2 provides the improvements to the physics fidelity through fine tuning. The full residual constructions including constraints are reported in Appendix A.1
2. **Out-of-distribution evaluation.** Yes. We evaluate GridSFM under topology-changing OPS inputs and on held-out N-1 AC-OPF cases. For the checkpoints adapted only on FullTop data, these tests introduce topology variation outside the system-specific adaptation distribution; Model 4 additionally uses N-1 adaptation to study whether topology-aware fine-tuning improves transfer to this regime. Figure 2 illustrates the representative J-S1 topology shift, Table 1 quantifies its effect on the released checkpoint, and Table 2 reports the common FullTop, N-1, intact, and modified evaluations across Models 1–4. Full evaluation details are provided in Appendix C.
3. **Failure modes.** Limited by our current scope and left for future work. This work serves as a controlled diagnostic study for the integration of GridSFM into the OPS problem. While we do report physics infeasibility as seen in Table 1, worst-case or failure modes analysis were not the focus of this study and thus not performed. However, we look to extend this work with security-constrained analysis across realistic topology and wildfire scenarios which naturally considers worst-case scenarios as part of the main focus.
4. **System-level effect.** Yes. A key focus in this paper was to consider the downstream effects of embedding GridSFM as an electrical-state evaluator within the larger OPS decision workflow. From Table 1 to Table 2, we assess how lightweight fine tuning reduces the training loss and subsequently the physics infeasibility as compared to the original model. With these improvements, we then investigated the potential impacts on the OPS evaluated behavior with Section 3.3 on the improvements to the evaluated frontiers in Figure 3 in the main paper and Figure 4 in the Appendix. Subsequent downstream investigations are covered in Section 3.3 alongside Appendix D with the Reference A/B and warm start studies.
5. **Data and code.** Will provide full package of data, code, and results upon acceptance. We utilize GridSFM-Open as the pre-trained foundation model alongside the open-source PGLib-OPF system to create our fine-tuning checkpoints. Thus, our work will be open-access around the time of the acceptance/rejection decision as we encourage others to build upon this study's work.